

StrataDream: Layer-wise 3D Gaussian Scene Reconstruction from a Single Image

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Paper ID #1195

Abstract. Recent advances in generative models have increased the demand for methods that can reconstruct high-quality 3D scenes from limited visual input. 3D Gaussian Splatting (3DGS) has become a powerful representation for real-time novel-view synthesis, and recent training-free 3DGS methods can reconstruct scenes without task-specific large-scale 3D training by directly optimizing a scene from the input. However, when the input is limited to a single image, foreground occlusions obscure background regions, making novel-view consistency difficult. Many non-layered training-free methods optimize the scene as a single Gaussian representation, which can leave foreground-occluded background regions unrecovered and produce unstable component boundaries under novel-view extrapolation. We propose **StrataDream**, a training-free layer-wise 3D Gaussian reconstruction framework for single-image 3D scene generation. StrataDream decomposes a single image into background, foreground, and sky layers, restores the background regions occluded by foreground objects, and reconstructs each layer as an independent 3D Gaussian scene with disjoint parameters, reducing interference among scene components during layer-wise reconstruction. The independently reconstructed layers are then merged into a unified 3D scene for novel-view synthesis. Experiments on the WorldScore benchmark show that StrataDream improves consistency-related metrics over baseline methods. Qualitative results further demonstrate that our layer-wise reconstruction reduces occlusion-induced holes and preserves cleaner boundaries between scene components.

Keywords: Layer-wise Gaussian Reconstruction · Single-image 3D Generation · 3D Gaussian Splatting

1 Introduction

Recent advances in generative models have substantially changed how visual content is produced and manipulated. Diffusion-based image generation [15, 27], video generation [3, 16], and image editing [4, 14, 24] have demonstrated remarkable visual quality and broad applicability across creative and industrial domains. This progress has naturally extended generative modeling from 2D visual content toward 3D scene construction and generation. High-quality 3D representations are increasingly required in AR/VR, robotics, autonomous driving, digital twins, and content creation [1, 7]. In particular, reconstructing a 3D scene

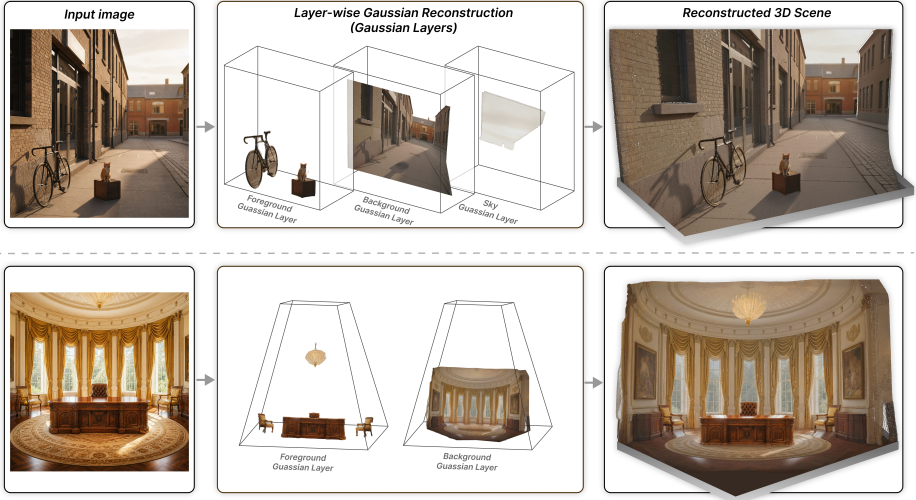


Fig. 1: Overview of StrataDream. StrataDream reconstructs a 3D Gaussian scene from a single image by decomposing the scene into foreground, background, and sky components. The components are reconstructed as independent Gaussian layers and then composed into a final 3D scene, enabling cleaner component boundaries and more consistent novel-view synthesis.

from only a small amount of visual input has become an important problem, as practical applications often require users to create 3D content from sparse views or even a single image [23, 35].

Among these settings, single-image 3D scene reconstruction is especially challenging because it is inherently ill-posed. Unlike multi-view reconstruction, a single image provides only one visible projection of the scene, leaving large portions of geometry, appearance, and background content unobserved. This difficulty becomes more pronounced in open-world scenes, where foreground objects, distant backgrounds, and sky regions exhibit different geometric properties, depth distributions, and visibility patterns. Therefore, a successful single-image reconstruction method should not only infer plausible unseen content, but also maintain stable geometry and appearance under novel-view extrapolation.

Neural Radiance Fields (NeRF) [25] have played a central role in neural 3D reconstruction by representing a scene as a continuous radiance field parametrized by a multi-layer perceptron. NeRF-based methods have demonstrated impressive novel-view synthesis quality, but they typically require long per-scene optimization times and expensive volumetric sampling during rendering, which limits their practicality for efficient high-resolution scene reconstruction [12, 22]. To address these limitations, recent research has increasingly adopted 3D Gaussian Splatting (3DGS) [19], which represents a scene as an explicit set of 3D Gaussians and renders novel views through differentiable splatting. Compared with NeRF, 3DGS provides faster optimization and real-time rendering while main-

taining high visual quality, making it a powerful representation for efficient 3D scene reconstruction.

Recent 3DGS-based reconstruction methods can be broadly categorized into training-based and training-free approaches. Training-based methods train feed-forward or generalizable models on large-scale 3D or multi-view datasets, enabling rapid reconstruction at inference time [6, 30, 34]. These methods can produce plausible 3D representations from sparse views or even limited visual input by leveraging learned 3D priors. However, their generalization performance depends heavily on the scale and diversity of the training data, and constructing datasets that sufficiently cover real-world scene variation remains difficult. In addition, training such models requires substantial computational resources and time.

In contrast, training-free methods do not require task-specific large-scale 3D training or fine-tuning. Instead, they directly optimize a 3D representation for a given input image or video, often with the help of off-the-shelf generative, depth estimation, segmentation, or inpainting priors [8, 29]. This makes them attractive for open-world single-image 3D scenes that may not belong to a well-defined training distribution. Because they optimize each scene individually, these methods can flexibly adapt to diverse visual content without requiring a dedicated 3D dataset.

Despite this flexibility, many non-layered training-free or per-scene 3DGS pipelines [8, 29] suffer from a fundamental coupling problem. These methods often initialize and optimize the entire scene as a single Gaussian representation, where foreground objects, background regions, and sky-like distant components are jointly explained by one set of parameters. This design leads to two recurring failure modes. First, a single input image provides no observation of the background hidden behind foreground objects. When the camera moves away from the input viewpoint, these unobserved regions are exposed as disocclusion holes, incomplete surfaces, or torn geometry. Second, because different scene components are represented within a single Gaussian space, Gaussians associated with one component can interfere with another, causing cross-component leakage, blurred silhouettes, and unstable foreground-background boundaries. These failures are therefore not merely artifacts of hyperparameter choices; rather, they arise from the coupled representation used by conventional single-view 3DGS pipelines.

A natural way to reduce this ambiguity is to decompose the scene into explicit layers. Recent layered scene generation methods, most notably WonderWorld [36], introduce fast LAYered Gaussian Surfels (FLAGS) and generate foreground, background, and sky-like layers from a single view, enabling interactive 3D scene generation with very low latency. WonderWorld further optimizes layers in a back-to-front manner by freezing previously optimized layers, making it highly effective for rapid scene prototyping. However, its primary goal is interactive world expansion rather than high-fidelity single-image 3D scene reconstruction. Its FLAGS representation is designed for speed through surfel-based geometry initialization and efficient refinement of selected surfel parameters.

As acknowledged in its limitations, WonderWorld is constrained by front-facing surfaces and may still produce holes or floaters for detailed objects under view-point changes [36]. This leaves room for a reconstruction-oriented framework that preserves the benefits of layer decomposition while reconstructing each component as an independent Gaussian scene for improved cross-view consistency and boundary stability.

We propose **StrataDream**, a training-free framework for single-image 3D scene reconstruction that performs layer-wise reconstruction at the Gaussian-scene level. Given a single image, StrataDream first decomposes the scene into foreground, background, and sky layers. It then explicitly completes the foreground-occluded background before 3D lifting, providing plausible content for regions that would otherwise appear as disocclusion holes under novel-view extrapolation. Unlike speed-oriented layered surfel generation, StrataDream lifts each layer into an independent 3D Gaussian scene with disjoint parameters and optimizes the layers through separate reconstruction paths. This design allows each component to preserve its own geometry and appearance while reducing cross-component contamination at object boundaries. The independently reconstructed layers are finally merged into a unified 3D representation. By combining occlusion-aware background completion, independent Gaussian-scene reconstruction, and final layer composition, StrataDream targets high-quality single-image 3D reconstruction without requiring task-specific 3D training or fine-tuning.

Experiments on the WorldScore benchmark [9] show that StrataDream improves single-image 3D scene reconstruction among the compared methods, particularly on text-independent consistency-related metrics. These results support our central claim that reconstruction-oriented independent Gaussian-scene optimization leads to more stable geometry and appearance under novel viewpoints. Qualitative comparisons further show that StrataDream produces fewer disocclusion holes and cleaner boundaries between scene components. In summary, our main contributions are as follows:

- We identify two structural challenges in single-image 3DGS reconstruction: foreground-induced background disocclusion and cross-component boundary instability among foreground, background, and sky regions.
- We propose **StrataDream**, a training-free single-image 3D scene reconstruction framework that decomposes a scene into foreground, background, and sky layers and reconstructs each layer as an independent 3D Gaussian scene with disjoint parameters.
- We introduce an occlusion-aware layer reconstruction and composition pipeline that completes foreground-occluded background regions before 3D lifting and integrates independently reconstructed layers into a unified scene, achieving cleaner component boundaries and improved cross-view consistency on WorldScore.

2 Related Work

2.1 Novel View Synthesis from a Single Image

Novel view synthesis from a single image aims to infer a 3D representation from a single observed view and render the scene from new camera poses. Early approaches explored explicit scene representations such as point clouds [26], multi-plane images [31], layered depth images [28], and meshes [18], which enable efficient rendering and plausible results for moderate viewpoint changes. However, because these representations are reconstructed mainly from the observed image content, they struggle to represent regions outside the input field of view or hidden by occluders. This limitation motivates recent scene-generation approaches, which expand the scene by hallucinating unobserved regions to support larger camera motions.

2.2 3D Scene Generation

Recent 3D scene generation methods expand a single image or text prompt into a larger 3D environment by combining camera movement, monocular depth estimation, image inpainting, and explicit 3D representations. SceneScape [11] progressively constructs a textured mesh for text-driven walkthrough generation, and Text2Room [17] generates room-scale textured meshes from text prompts. WonderJourney [37] generates diverse connected scenes with language and vision-language guidance, InvisibleStitch [10] improves depth coherence through depth inpainting, and LucidDreamer [8] optimizes 3D Gaussian scenes from generated multi-view content. These methods can synthesize large, visually plausible scenes, but many of them extend the scene along a camera path without explicitly separating the foreground, background, and sky components. As a result, background regions hidden behind foreground objects in the input view can remain unrecovered, leading to holes, incomplete geometry, or unstable component boundaries under novel viewpoints.

To mitigate foreground-induced background occlusion, recent layered scene generation methods explicitly separate scene components. WonderWorld [36] introduces FLAGS with foreground, background, and sky layers, together with layer-level inpainting for fast interactive world generation. This layered design helps recover content hidden by foreground objects and supports efficient scene expansion. However, WonderWorld is primarily designed for low-latency scene prototyping rather than reconstruction-oriented layer-wise Gaussian optimization. Its speed-oriented representation and lightweight refinement make interactive generation efficient, but preserving fine-grained object geometry and stable component boundaries under viewpoint changes remains challenging [36].

In contrast, StrataDream targets reconstruction-oriented single-image 3D scene generation. Rather than refining speed-oriented surfels, StrataDream reconstructs foreground, background, and sky as independent 3D Gaussian scenes with disjoint parameters. It also explicitly restores foreground-occluded background regions before 3D lifting, so that hidden background content can be

reconstructed as part of the background Gaussian scene. This design prioritizes layer-specific geometry and cross-view consistency over interactive speed. We detail this reconstruction pipeline in Section 3.

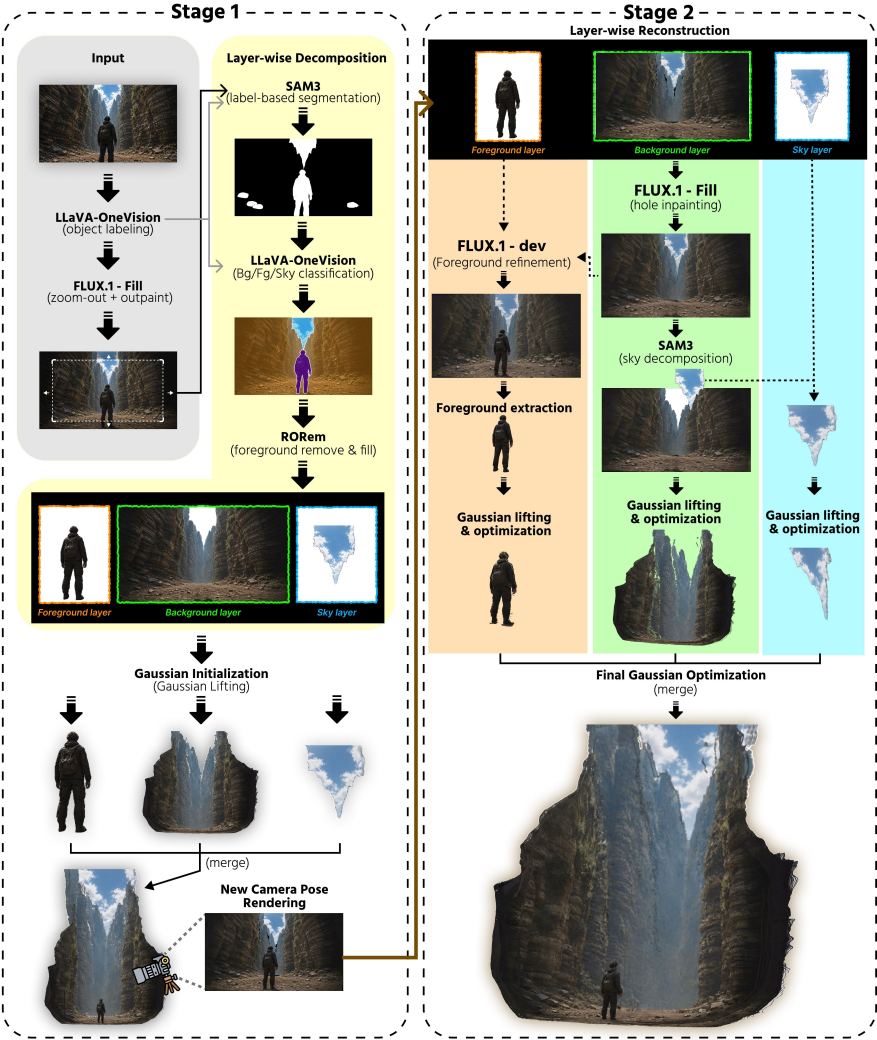


Fig. 2: StrataDream pipeline. Our pipeline reconstructs a 3D Gaussian scene from a single image in two stages. Stage 1 decomposes the input into foreground, background, and sky layers and recovers the background hidden behind foreground objects before Gaussian initialization. Stage 2 progressively reconstructs each layer along a synthetic camera trajectory through layer-wise updates and Gaussian optimization. The reconstructed layers are then merged and refined into a single Gaussian scene for novel-view synthesis.

3 Method

3.1 Overview

Given a single-view image I_0 , StrataDream reconstructs a 3D Gaussian scene for novel-view synthesis. As illustrated in Figure 2, our pipeline consists of two stages. In Stage 1, we decompose the input image into background, foreground, and sky layers, recover the background hidden behind foreground objects, and initialize layer-wise Gaussian scenes. In Stage 2, we progressively update each layer along a synthetic camera trajectory through layer-specific inpainting, lifting, and optimization. The independently reconstructed layers are finally merged and refined into a single 3D Gaussian scene.

3.2 Occlusion-aware Layer Decomposition

We first construct aligned background, foreground, and sky layers from the input image. We use LLaVA-OneVision [20] to obtain candidate object labels. To enlarge the field of view, we then place the input image at the center of a zoomed-out canvas and complete the newly introduced boundary regions using FLUX.1-Fill [2], producing an outpainted image.

The candidate object labels serve as text prompts for SAM 3 [5] to obtain per-instance masks on the outpainted image. Since the masks alone do not determine the scene component of each object, we classify the labels into foreground, background, and sky categories using LLaVA-OneVision and merge the masks accordingly, yielding semantic masks M^{fg} , M^{bg} , and M^{sky} for the three components. To recover the background occluded by foreground objects, we remove the foreground regions from the outpainted image using RORem [21], which fills them with plausible background content. The foreground layer is extracted from the outpainted image using M^{fg} , and the sky layer from the foreground-removed image using M^{sky} .

The background layer, whose color image L^{bg} is taken from the foreground-removed image, is defined over the entire non-sky region rather than only the semantic background M^{bg} : the pixels originally covered by the foreground now carry recovered background appearance and must also be represented in 3D so that they do not appear as holes under novel views. We therefore denote the *background support* by $M^{\text{non-sky}}$, the non-sky region spanning both the semantic background and the recovered foreground footprint, i.e. $M^{\text{non-sky}} = M^{\text{bg}} \cup M^{\text{fg}}$. These aligned layers and masks are passed to the next stage for layer-wise Gaussian reconstruction.

3.3 Layer-wise Gaussian Initialization

For each layer $l \in \{\text{bg}, \text{fg}, \text{sky}\}$, let L^l , D^l , and M^l denote its color image, depth map, and valid mask, respectively. Given camera parameters $\Pi = (K, R, \mathbf{t})$, we lift each layer into a Gaussian scene using

$$\mathcal{G}^l = \Phi(L^l, D^l, M^l, \Pi) = \{g(\mathbf{X}^l(\mathbf{p}), L^l(\mathbf{p})) : \mathbf{p} \in M^l\}, \quad (1)$$

where $\mathbf{X}^l(\mathbf{p})$ is obtained by back-projecting pixel $\mathbf{p}$ with depth $D^l(\mathbf{p})$, and $g(\mathbf{X}, c)$ instantiates a Gaussian centered at $\mathbf{X}$ with color c .

For the background and foreground layers, we estimate metric depth using MoGe-2 [32]. Since the foreground and background are reconstructed along separate paths, we align the foreground depth to the background depth in inverse-depth space using a per-image affine fit:

$$\mathcal{A}(D, D_{\text{ref}}; \Omega) = \left(\frac{a^*}{D} + b^* \right)^{-1}, \quad (2)$$

where (a^*, b^*) is obtained by least-squares fitting $a/D + b$ to $1/D_{\text{ref}}$ over Ω . We set $\hat{D}^{\text{fg}} = \mathcal{A}(D^{\text{fg}}, D^{\text{bg}}; \Omega_{\text{fg}})$ over the overlap region Ω_{fg} where the foreground and background layers have valid depth, placing the foreground Gaussians at a consistent relative depth with respect to the background. For sky regions, we do not rely on monocular depth estimation; instead, we place sky Gaussians on a fixed far sphere to maintain a stable sky boundary across views.

The initial layer-wise Gaussian scenes are obtained at the input view Π_0 . The background is lifted over its support $M^{\text{non-sky}}$, including the recovered regions behind the foreground, while the foreground is lifted over its mask M^{fg} :

$$\mathcal{G}_0^{\text{bg}} = \Phi(L^{\text{bg}}, D^{\text{bg}}, M^{\text{non-sky}}, \Pi_0), \quad \mathcal{G}_0^{\text{fg}} = \Phi(L^{\text{fg}}, \hat{D}^{\text{fg}}, M^{\text{fg}}, \Pi_0). \quad (3)$$

When sky regions exist, $\mathcal{G}_0^{\text{sky}}$ is initialized by placing Gaussians on the fixed sky sphere.

3.4 Progressive Layer-wise Reconstruction

Starting from the initialized Gaussian scenes, we progressively reconstruct the scene along a synthetic camera trajectory. At each pose Π_t , we render the layer-composited scene $\mathcal{G}_t^{\text{comp}} = \mathcal{G}_t^{\text{bg}} \cup \mathcal{G}_t^{\text{fg}} \cup \mathcal{G}_t^{\text{sky}}$, which is used only for rendering and hole detection. Pixels whose accumulated alpha is lower than a threshold τ_α form the missing region to be completed.

Each layer is then updated along its own reconstruction path. For the background path, foreground regions are removed from the rendered view, and missing background regions are completed using FLUX.1-Fill. For the foreground path, the foreground region is refined using FLUX.1-dev [2], and its silhouette $M_{\text{new}}^{\text{fg}}$ is re-extracted using SAM 3. Outside $M_{\text{new}}^{\text{fg}}$, the refined view is reverted to the background-path result to provide surrounding context for foreground depth estimation, while only the content within $M_{\text{new}}^{\text{fg}}$ is lifted into the foreground scene. For the sky path, sky pixels are lifted onto the fixed sky sphere.

For depth-estimated layers, we align the newly predicted depth to the rendered depth of the same layer before lifting:

$$\hat{D}^l = \mathcal{A}(D_{\text{pred}}^l, D_r^l; \Omega_t^l), \quad l \in \{\text{bg}, \text{fg}\}, \quad (4)$$

where D_{pred}^l is the depth predicted for the newly generated view, D_r^l is the depth rendered from the current Gaussian scene of the same layer, and Ω_t^l is the overlap

region already covered by that layer. After obtaining the updated layer image $\hat{L}^l$, we add new Gaussians by lifting the full valid region of the layer:

$$\mathcal{G}_t^l \leftarrow \mathcal{G}_{t-1}^l \cup \mathcal{O}_l \left(\Phi(\hat{L}^l, \hat{D}^l, M^l, \Pi_t) \right). \quad (5)$$

Here, the background is lifted over its non-sky support $M^{\text{bg}} = M^{\text{non-sky}}$, the foreground over its per-pose silhouette $M^{\text{fg}} = M_{\text{new}}^{\text{fg}}$, and the sky over M^{sky} . The occlusion filter $\mathcal{O}_l$ discards a lifted Gaussian if it projects in front of already-reconstructed geometry at any processed pose; it is applied to $l \in \{\text{bg}, \text{fg}\}$ and becomes the identity for the sky layer. Although inpainting is applied only to missing regions, lifting is performed over the full valid layer region. This allows recovered background content to be inserted into the background Gaussian scene rather than discarded.

3.5 Separate-path Layer Optimization

Unlike methods that optimize all Gaussians together in a single Gaussian scene, StrataDream keeps background, foreground, and sky as separate Gaussian scenes throughout progressive reconstruction. Because the layers have disjoint parameters, Gaussians in one layer are not updated by supervision from another layer during layer-wise optimization. This reduces direct cross-component updates, allowing each layer to better preserve its own geometry and appearance before the final refinement stage.

At pose t , each layer maintains an accumulated target set $\mathcal{D}_t^l = \{(I_i^l, M_i^l, \Pi_i)\}_{i=0}^t$, where I_i^l , M_i^l , and Π_i are the target image, valid mask, and camera pose at pose i . During optimization, target views are sampled from $\mathcal{D}_t^l$, and only the newly added Gaussians are optimized while previously reconstructed Gaussians are kept as anchors. This constrains new Gaussians using both the current and previously generated views without overwriting earlier reconstruction.

For a target view $(I_{\text{gt}}^l, M^l, \Pi_i)$ sampled from $\mathcal{D}_t^l$, we render layer l as $I_r^l = \mathcal{R}(\mathcal{G}_t^l, \Pi_i)$ and minimize the masked reconstruction loss

$$\mathcal{L}_l = \text{SmoothL1} \left(I_r^l[M^l], I_{\text{gt}}^l[M^l] \right) + \lambda_{\text{ssim}} \left(1 - \text{SSIM}(I_r^l, I_{\text{gt}}^l) \right). \quad (6)$$

The SmoothL1 [13] term is restricted to the valid region of each layer by the mask, and λ_{ssim} balances the SSIM [33] term. The accumulated target set further encourages consistency with previously generated views.

3.6 Scene Composition and Final Refinement

After each camera pose is processed, the layer-wise Gaussian scenes are composed to render the next pose, while each layer remains in its own separate optimization path. Once all camera poses have been processed, we merge the independently reconstructed layers into a final Gaussian scene:

$$\mathcal{G}^{\text{final}} = \text{copy} \left(\mathcal{G}^{\text{bg}} \cup \mathcal{G}^{\text{fg}} \cup \mathcal{G}^{\text{sky}} \right), \quad (7)$$

where $\text{copy}(\cdot)$ duplicates the layer-wise Gaussians into a single scene so that joint refinement does not overwrite the separately reconstructed layers. We then jointly refine all Gaussians under full-image supervision using the original input view as a strong anchor and the generated views as additional constraints. This final refinement improves the coherence of the final Gaussian scene while preserving the layer-wise reconstruction obtained from the separate paths. The optimized final Gaussian scene is used for novel-view synthesis.

Table 1: Quantitative comparison on the WorldScore benchmark [9].

Since our method does not rely on text prompts, we report five text-independent metrics—camera controllability (Camera), 3D consistency (3D), photometric consistency (Photo), style consistency (Style), and subjective quality (Subj.)—along with their average (Avg). All metrics are normalized such that higher is better. Baseline scores are taken directly from WorldScore [9] and averaged over these same five metrics. For each column, the best result is in **bold** and the second best is underlined.

Method	Camera	3D	Photo	Style	Subj.	Avg
SceneScape [11]	84.99	76.54	62.88	21.85	32.75	55.80
Text2Room [17]	94.01	88.71	88.36	37.23	36.69	69.00
LucidDreamer [8]	88.93	<u>90.37</u>	<u>90.20</u>	48.10	<u>58.99</u>	75.32
WonderJourney [37]	84.60	80.60	79.03	62.82	66.56	74.72
InvisibleStitch [10]	<u>93.20</u>	88.51	89.19	32.37	58.50	72.35
WonderWorld [36]	92.98	86.87	85.56	<u>70.57</u>	49.81	<u>77.16</u>
StrataDream (Ours)	87.92	90.55	92.33	91.93	33.90	79.32

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate StrataDream on the static subset of the WorldScore benchmark [9], which contains 2000 examples (1000 photorealistic and 1000 stylized) spanning indoor and outdoor scenes. Each example provides a single reference image and a predefined camera trajectory. We take the reference image as the single-view input and render the reconstructed scene along the provided trajectory for evaluation.

Baselines. We compare against the six 3D scene generation methods in WorldScore [9]: SceneScape [11], Text2Room [17], LucidDreamer [8], WonderJourney [37], InvisibleStitch [10], and WonderWorld [36]. Since all six are evaluated under the same WorldScore protocol, we report their scores directly from the benchmark.

Metrics. WorldScore [9] measures a set of metrics grouped into controllability, quality, and dynamics. Two controllability metrics, object controllability and content alignment, require a next-scene text prompt and thus do not apply to

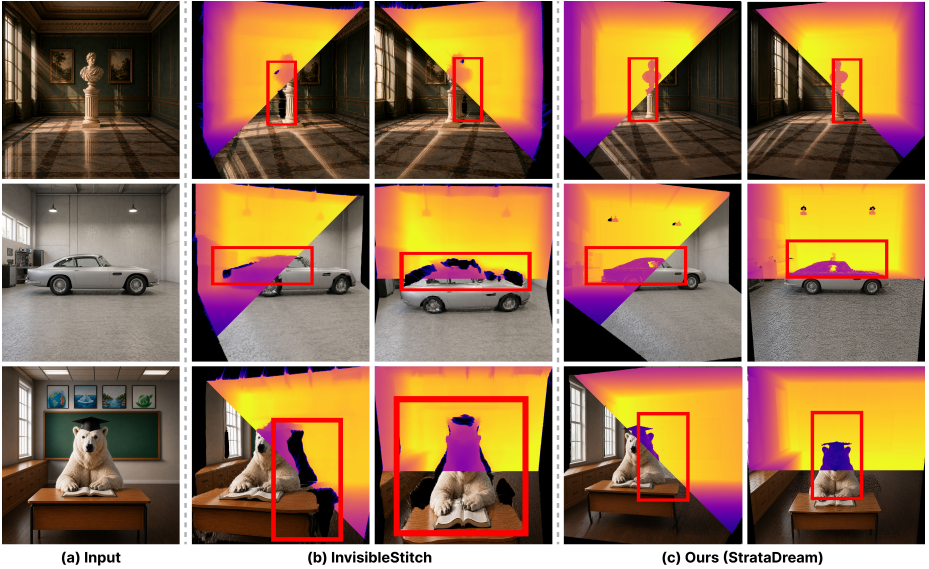


Fig. 3: Qualitative comparison with InvisibleStitch [10]. Given (a) a single input image, we compare novel-view renderings from (b) InvisibleStitch and (c) StrataDream. Rendered images are shown together with their corresponding depth maps. Red boxes highlight regions around foreground objects where previously occluded background content becomes visible under viewpoint changes. InvisibleStitch often produces holes or distorted geometry in these regions, whereas StrataDream shows fewer disocclusion artifacts by recovering foreground-occluded background content before 3D lifting.

our text-free single-image setting. We therefore report the five text-independent metrics—camera controllability, 3D consistency, photometric consistency, style consistency, and subjective quality—together with their average. For the baselines, we average their scores over the same five metrics to ensure a consistent comparison.

Implementation details. We conduct all experiments on a single RTX PRO 6000 Blackwell GPU, and reconstructing a scene takes roughly 10 to 15 minutes. All views are rendered at 512×512 , following the WorldScore camera trajectories for quantitative evaluation and a smooth trajectory over separate single-image inputs for the qualitative comparisons, where each baseline is run with its official implementation. Further details, including all hyperparameters and camera trajectories, are in the supplementary material.

4.2 Comparison with Baselines

As reported in Table 1, StrataDream is most effective on the consistency-oriented metrics of the WorldScore benchmark. It attains the best photometric consistency and style consistency, and is on par with the strongest baseline on 3D

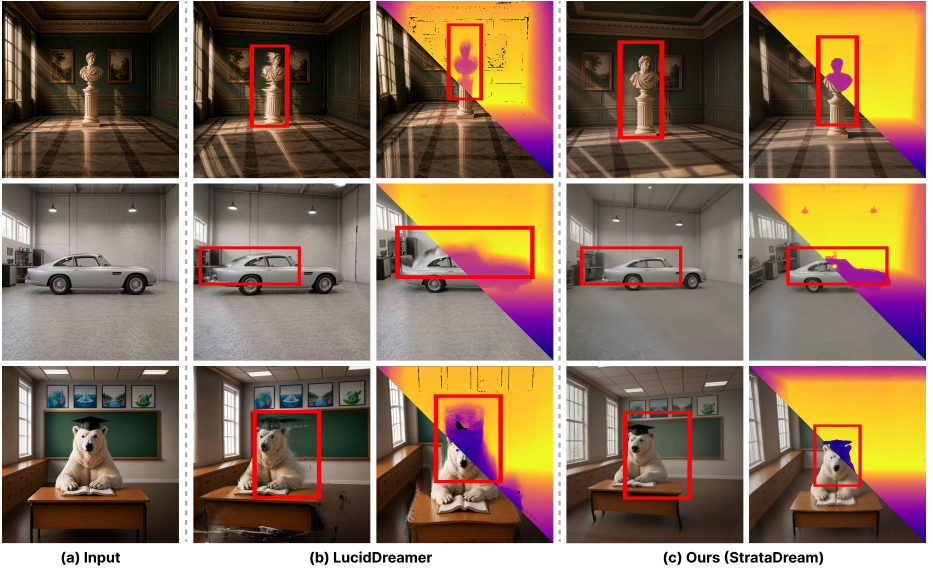


Fig. 4: Qualitative comparison with LucidDreamer [8]. Given (a) a single input image, we compare novel-view renderings from (b) LucidDreamer and (c) StrataDream. Rendered images are shown together with their corresponding depth maps. Red boxes highlight foreground-background boundary regions. LucidDreamer often shows blurred silhouettes, distorted geometry, or fragmented depth around foreground objects, whereas StrataDream preserves clearer object boundaries and more consistent depth through layer-wise Gaussian reconstruction.

consistency, with only a marginal 0.18-point lead over LucidDreamer. Because WorldScore measures 3D consistency in terms of geometric stability and photometric consistency in terms of cross-view appearance stability [9], these results indicate that the reconstructed scene preserves stable geometry and appearance as the camera moves away from the input view. This behavior is consistent with our objective of recovering foreground-occluded background regions and reconstructing background, foreground, and sky as separate Gaussian scenes.

WorldScore measures style consistency by comparing Gram matrices between the first and last frames, capturing global stylistic drift such as color shifts and changes in texture statistics [9]. This metric aligns well with the reconstruction-oriented design of StrataDream. Because our method anchors each layer to the input image and inpaints only disoccluded regions rather than re-synthesizing entire frames, it preserves stable color and texture statistics along the camera trajectory, leading to a substantial improvement in style consistency. At the same time, this design also relates to why StrataDream does not lead in subjective quality: by prioritizing layer-wise geometric and appearance consistency over full-frame generative enhancement, the rendered views may be less visually refined than those produced by methods that rely more heavily on generative



Fig. 5: Qualitative comparison with WonderWorld [36]. Given (a) a single input image, we compare novel-view renderings from (b) WonderWorld and (c) StrataDream. Rendered images are shown together with their corresponding depth maps. Red boxes highlight foreground objects and their adjacent background regions. While WonderWorld can produce plausible novel views, its depth maps show unstable component boundaries around foreground objects, whereas StrataDream produces cleaner foreground-background boundaries and more consistent depth.

re-synthesis. These per-metric results indicate that StrataDream is particularly effective for consistency-oriented reconstruction, while improving perceptual realism remains an important direction for future work.

To examine whether the quantitative gains in Table 1 correspond to visible improvements in reconstruction quality, we compare StrataDream with three representative baselines that exhibit different failure modes.

Figure 3 examines the effect of explicitly recovering foreground-occluded background regions. InvisibleStitch [10] expands the scene by connecting newly generated content to the existing geometry, but it does not explicitly complete the background hidden behind foreground objects before 3D lifting. As the camera moves away from the input view, these unobserved regions become visible and appear as holes or distorted geometry in the rendered depth maps, as highlighted by the red boxes. StrataDream instead removes foreground objects and completes the occluded background before 3D lifting, so the recovered content is reconstructed as part of the background Gaussian scene. Consistent with this design, StrataDream exhibits fewer disocclusion artifacts and more coherent background geometry under novel viewpoints.

Figure 4 illustrates the benefit of independent layer-wise reconstruction. LucidDreamer [8] reconstructs the scene as a single Gaussian scene, without

maintaining separate reconstruction paths for foreground, background, and sky. When foreground and background are represented within one scene, component interactions can lead to blurred silhouettes, unstable foreground–background boundaries, and fragmented depth around object contours. By reconstructing each component as an independent Gaussian scene with disjoint parameters, StrataDream preserves cleaner object boundaries and reduces depth artifacts around foreground regions.

Figure 5 compares StrataDream with WonderWorld [36], a recent layered scene generation method designed for fast interactive world expansion. WonderWorld benefits from explicit foreground/background/sky decomposition and low-latency layered generation, but its speed-oriented FLAGS representation can still leave fragmented depth or less stable boundaries for detailed foreground objects under viewpoint changes. StrataDream instead targets reconstruction-oriented layer-wise Gaussian reconstruction: by recovering foreground-occluded background regions and reconstructing each component as an independent 3D Gaussian scene, it produces cleaner component boundaries and more consistent depth in novel views.

5 Conclusion

We present StrataDream, a training-free framework for single-image 3D scene reconstruction based on layer-wise 3D Gaussian reconstruction. StrataDream decomposes an input image into background, foreground, and sky layers, restores plausible background content occluded by foreground objects, and reconstructs each layer as an independent 3D Gaussian scene before merging them into a final scene. This design addresses two major challenges in single-image 3DGS: disocclusion holes from hidden background regions and boundary instability from cross-component interference. Experiments on the WorldScore benchmark show that StrataDream improves consistency-oriented metrics, and qualitative comparisons demonstrate fewer occlusion-induced holes, cleaner component boundaries, and more coherent depth under novel viewpoints. These results suggest that reconstruction-oriented layer-wise Gaussian optimization is an effective strategy for improving cross-view consistency in single-image 3D scene reconstruction.

6 Future Work

Currently, StrataDream decomposes a scene into three fixed layers: background, foreground, and sky. Extending this formulation to an arbitrary number of layers would broaden its applicability to complex multi-depth scenes involving severe occlusions, dense foreground structures, and diverse object arrangements. Another promising direction is to extend StrataDream beyond static 3D reconstruction to dynamic 4D scene generation, where temporal changes must also be modeled consistently.

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